STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 1: Units and Measurements

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. How many SI base quantities?
2. SI unit of temperature?
3. Significant figures in 0.00450?
4. Dimensions of speed?
5. Dimensions of force?
6. Is length/length dimensionless?
7. Can dimensional analysis determine pi?
8. SI unit of pressure?

Section B 2-mark questions - 6 x 2 = 12 marks

9. Round 12.486 to 3 significant figures.
10. Dimensions of momentum mv?
11. Check s=ut+at^2 dimensionally.
12. Convert 72 km/h to m/s.
13. Significant figures in 5.070 x 10^3?
14. Why can’t dimensions distinguish work and torque?

Section C 3-mark questions - 4 x 3 = 12 marks

15. Derive pendulum period dependence T proportional to lg.
16. Sphere radius 2.00 cm: volume to 3 s.f.?
17. Dimensions of G from F=Gm1m2/r^2?
18. 15.2 cm x 3.4 cm: area with correct precision?

Section D 4-mark questions - 2 x 4 = 8 marks

19. Spring-mass period T proportional to mk, [k]=MT-^2. Find exponents.
20. Calculated volume is 18.237 cm^3, least precise input 3 s.f. What is reported?